CS & DA

Probability and Statistics

Permutations and Combinations

Lecture No. 03







By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

1. Permutation
2. Combination
3. Difference between P & C

Topics to be Covered



Topic

BASICS of P & C (Part 3)

Perm & Combination & (Counting Principle)

Fundamental Principle of Addition → if we have to perform only one of the job at a time out of m jobs then use this principle. Keywords: "Either or / only one / Anyone"

Fundamental Principle of Multiplication → if we have to perform all the jobs at a time (out of m jobs) then use this Principle

Keywords: "AND / Both / All"

Conclusion:- Let n = No. of places, r = No. of persons

① if $n > r$ & RNA then Multi Rule = Perm Rule

② if $n = r$ & RNA then Multi Rule = Perm-Rule = Factorial Rule

③ if RA then only use Multi Rule.

is the concept of ${}^nC_r$, ${}^nP_r$ & $r!$ exist only when R.N.A

Combination → (when ever Counting is Based on only selection then we will use Combination Rule)

$$^nC_r = \frac{n!}{r!(n-r)!}$$ = selection of r things from n things

Note: we know that $^nC_r = {}^nC_{n-r}$ for eg $^{22}C_{18} = ? = {}^{22}C_{22-18}$

$= {}^{22}C_4 = \frac{22\times21\times20\times19}{4\times3\times2\times1}$

shortcut: $^{11}C_3 = \frac{11\times10\times9}{3\times2\times1}$

$^{22}C_4 = \frac{22\times21\times20\times19}{4\times3\times2\times}$

eg $^7C_2 = \frac{7\times6}{2\times1}$, $^6C_4 = 6$, $^7C_4 = 7$

eg $^5C_1 = \frac{5}{1} = 5$, $^6C_0 = 1$, $^7C_0 = 1$

$^nC_0 = 1$, $^nC_1 = n$, $^nC_2 = \frac{n(n-1)}{2\times1}$, $^nC_n = {}^nC_0 = 1$, $^nC_0 = \frac{n!}{0!(n-0)!} = \frac{n!}{n!} = 1$

Permutation → when Counting is Based on Selection & Arrangement Both

(then we will use Perm. Rule)

→ Perm = Selection & Arrangement

$$^nP_r = \frac{n!}{(n-r)!} = {}^nC_r \times r!$$

^nC_r → Selection, $r!$ → Arrangement.

eg $^{11}P_3 = 11\times10\times9$, $^{22}P_4 = 22\times21\times20\times19$

$^{11}C_3 = \frac{11\times10\times9}{3\times2\times1}$

$^nP_0 = {}^nC_0 = 1$, $^nP_1 = n = {}^nC_1$

$^nP_n = n!$, $^nC_n = 1$,

$^nP_2 = n(n-1)$, $^nC_2 = \frac{n(n-1)}{2\times1}$

$^7P_2 = 7\times6$, $^7P_1 = 7$

$^7C_2 = \frac{7\times6}{2\times1}$, $^7C_1 = \frac{7}{1} = 7$

$^7C_0 = 1 = {}^7C_0$

Analysis:- 6G & 4L & Committee of 5 Persons is to be formed.

Case I	Case II	Case III	Case IV	Case V	
(0L & 5G)	or (1L & 4G)	or (2L & 3G)	or (3L & 2G)	or (4L & 1G)	= Total Committees
$({}^4C_0 \times {}^6C_5)$	$+ ({}^4C_1 \times {}^6C_4)$	$+ ({}^4C_2 \times {}^6C_3)$	$+ ({}^4C_3 \times {}^6C_2)$	$+ ({}^4C_4 \times {}^6C_1)$	$= {}^{10}C_5$

$(1 \times 6) + (4 \times 15) + (6 \times 20) + (4 \times 15) + (1 \times 6) = 252$

$6 + 60 + 120 + 60 + 6 = 252$

At Most 2L

At least 3L

At least 2L

At least one L

Concept of At least and at Most →

eg: If we want to distribute 5 Apples among Children then various possibilities are as follows:

0 App or 1 App or 2 App or 3 App or 4 App or 5 App = Total Cases

At most 1 App

At most 2 App

At least 2 App.

At least 3 app.

At least one App

∴ At least one = Total − None

Q. Three dices are rolled simultaneously then find the number of outcomes in which at least one die show digit 4 ?

Sol: Total possible outcomes $= \underset{D_1}{6} \times \underset{D_2}{6} \times \underset{D_3}{6} = 216$ outcomes

No. of outcomes in which No Die show digit '4' $= ? = \underset{D_1}{5} \times \underset{D_2}{5} \times \underset{D_3}{5} = 125$ outcomes

Hence No of outcomes in which at least one die show digit 4

$=$ Total outcomes $-$ No dice show digit 4

$= 216 - 125 = 91$ outcomes.

Various outcomes = $\{(111)(112)(113)(114)(115)(116)$
$(121)(122)(123)\ldots\ldots(126)$
$(131)(132)(133)\ldots\ldots(136)$
$\vdots$
$(611)(612)(613)\ldots\ldots(616)$
$\ldots\ldots\ldots\ldots\ldots(666)\}$

$= \underset{D_1}{6} \times \underset{D_2}{6} \times \underset{D_3}{6}$

$= 216$ outcomes

No. of outcomes in which No Die show digit '4' = ?

$= \{(111)(112)(113)(114)(115)(116)$
$(121)(122)(123)(124)(125)(126)$
$(123)\ldots\ldots$
$(211)(212)(213)(214)(215)(216)$
$\ldots\ldots\ldots(614),(641)(666)\}$

$= 5 \times 5 \times 5 = 125$

Q: How many Nos are there b/n 99 & 1000 in which at least one digit is 7 ?

is we will consider only 3 digit Nos

Sol: Total 3 digit Nos that can be formed (RA) $= \underset{P_1}{9 \text{ ways}} \times \underset{P_2}{10 \text{ ways}} \times \underset{P_3}{10 \text{ ways}} = 900$ Nos

(Digits are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

Total 3 digit Nos in which No digit is '7' $= ? = \underset{P_1}{8 \text{ ways}} \times \underset{P_2}{9 \text{ ways}} \times \underset{P_3}{9 \text{ ways}} = 648$

Hence total 3 digit Nos in which at least one digit is 7

$=$ Total $-$ No Digit is 7 $= 900 - 648 =$ 252

(*) Number of linear Arrangements of n different things $= n!$

(*) Number linear Arrangements of n things in which p are alike, q are alike, r things are alike & Rest are different =

$$= \frac{{}^{n}P_{n}}{p!q!r!} = \frac{n!}{p!q!r!}$$

where $p+q+r+\underset{(?)}{\text{Rest}} = \text{Total}(n)$

Eg: No. of linear arrangements of a, b, c, d w/o Repetition = ? $= 4! = 24$

(RNA)

Eg No. of linear arrangements of a, a, a, b = ? (MI) $= \frac{4!}{3!} = 4$

(MI) Various arrangements are (aaab), (aaba), (abaa), (baaa) i.e Ans = 4

Eg No of linear arrangements of aa, b, b = ? (MI) $= \frac{4!}{2!\,2!} = 6$

(MII) Various Arrangements are (aabb), (abab), (abba), (bbaa), (baba), (baab) i.e Ans = 6

Q: How many different 4 digit Nos can be formed using the digits 5, 5, 6, 6

M-I Various Nos are (5566), (5656), (5665), (6655), (6565), (6556)

∴ Ans is 6 Nos can be formed using above Digits

M-II Req Ans $= \frac{4!}{2!\,2!} = 6$

(RNA)

Note: (RA) ⟹ we are free to use any digit, desired number of times

(2) when in a Question, digits or letters are given then we will take the Concept as (RNA) in general.

Q: How many different 9 letter words can be formed using the letters of the word ALLAHABAD

given letters are A, A, A, A, L, L, H, B, D

p = 4, q = 2 & Total letters $n = 9$

So Req Ans $= \frac{n!}{p!\,q!} = \frac{9!}{4!\,2!}$

Q: How many different 11 letter words can be formed using the letters of the word MATHEMATICS ?

MM, AA, TT, E, H, I, C, S

$p = 2$, $q = 2$, $r = 2$

$n = 11$ $\quad$ Ans $= \frac{11!}{2!\,2!\,2!}$

Q: In how many ways 2 Maths Books, 3 Physics Books, 4 Chemistry Books, 1 Eng Book, 1 Hindi Book & 1 Geography Book can be arranged in a shelf.

Sol:

$\underbrace{M, M}_{p=2}, \underbrace{P, P, P}_{q=3}, \underbrace{C, C, C, C}_{r=4}, H, E, G$ & Total items $(n) = 12$

So Req No of Arrangements $= \dfrac{12!}{2!\,3!\,4!}$

HW8 There are 6 periods in each working day of a school then in how many ways 5 different subjects can be arranged so that each subject gets at least one period? (Ans = 1800)

Sol: Subjects : P, C, E, B, M

Periods : I, II, III, IV, V, VI

eg: P C E B M E

I II III IV V VI

$$\text{Req Ans} = {}^{5}C_{1} \times \frac{6!}{2!} = 1800$$

$$= 5 \times \frac{6!}{2!} = 1800$$

Analysis: Various Cases :-

= (PCEMBP) or (PCEMBC) or (PCEMBE) or (PCEMBM) or (PCEMBB)

$= \frac{6!}{2!} + \frac{6!}{2!} + \frac{6!}{2!} + \frac{6!}{2!} + \frac{6!}{2!} = 5 \times \frac{6!}{2!}$

$= 360 + 360 + 360 + 360 + 360$

$= 360[1+1+1+1+1]$

$= 360 \times 5$

$= 5 \times 360$

$= 5 \times 360$

$= 1800$

Derrangements (All wrong arrangements) →

when No one goes at Right place assigned for him then such types of arrangements are called De Arrangements.

for n different things, Total No of Arrangements $= n!$

All Correct " $= 1$

& All wrong arrangements $= n!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\cdots\cdots(-1)^n\frac{1}{n!}\right]$

this formula is applicable when (RNA)

eg for $n=2$, No. of De-arrangements $= 2!\left[1-\frac{1}{1!}+\frac{1}{2!}\right] = 2\left[0+\frac{1}{2}\right] = 1$

eg for $n=3$, " " " $= 3!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}\right] = 6\left[0+\frac{1}{2}-\frac{1}{6}\right] = 2$

eg for $n=4$, " " " $= 4!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right] = 24\left[0+\frac{1}{2}-\frac{1}{6}+\frac{1}{24}\right] = 9$

eg for $n=5$, " " " $= 5!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right] = \cdots = 44$

eg for $n=6$, " " " $= 6!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}\right] = \cdots = 265$

eg: If there are 3 letters and 3 addressed Envelopes then Various arrangements are ?

Sol: Total arrangements $= 3! = 6$ (RNA)

, L_1, L_2, L_3
E_1, E_2, E_3

Various Arrangements are;

$\begin{pmatrix} E_1 & E_2 & E_3 \\ L_1 & L_2 & L_3 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_1 & L_3 & L_2 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_2 & L_1 & L_3 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_2 & L_3 & L_1 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_3 & L_1 & L_2 \end{pmatrix}, \begin{pmatrix} E_1 & E_2 & E_3 \\ L_3 & L_2 & L_1 \end{pmatrix}$

All correct Arrangements = 1

All wrong Arrangements = 2

Q: Six students are appearing in a test & their are six directed chairs then Find the Number of arrangements in which No student will be seated on his/her chair?

Dearrangement.

Sol: for $n=6$, Total Seating arrangements $= 6! = 720$

(RNA)

All Correct arrangements $= 1$

& All wrong arrangements $= 6!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}\right] = 265$ Ans

(ii) Find the No. of arrangements in which at least 2 Students will be seated on wrong chair = ?

(a) 720 (b) 265 (c) 1 (d) 719

At least 2 on a wrong chair = Total − No wrong

= Total − all Correct = 720 − 1 = 719

Various possibilities

All Correct.

(No wrong) or (2 wrong) or (3 wrong) or (4 wrong) or (5 wrong) or (6 wrong) = Total

All wrong = Derangements

At least 2 wrong

$= 6!$

$${}^6C_0 \times 1 + {}^6C_2 \times 1 + {}^6C_3 \times 2 + {}^6C_4 \times 9 + {}^6C_5 \times 44 + {}^6C_6 \times 265 = 720$$

$$1 + (15 \times 1) + (20 \times 2) + (15 \times 9) + (6 \times 44) + (1 \times 265) = 720$$:)

Q: (CS 2004) There are 5 Balls and 5 distinct Cells then Find the Number of arrangements in which Ball B_i is not placed in Cell C_i $\forall i$ & No Cell remains empty ?

RHA

De-arrangements = All wrong arrangements

Sol:-

For $n=5$, No. of De-Arrangements $= 5!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right] = 44$ Ans

Note:-
1. Total arrangements $= 5! = 120$
2. All Correct arrangements $= 1$

Note: ③ Above 44 De arrangements do not include following types of De-Arrangements



So Many De-Arrangements are possible of above types.

Tel: drpuneet@rpw

THANK - YOU